

PHYSICAL SCIENCES

PHYSICAL SCIENCES GRADE 12 TERM 3 CONTROLLED TEST

SUBJECT	PHYSICAL SCIENCES
GRADE	12
ASSESSMENT TASK	PHYSICAL SCIENCES GRADE 12 TERM 3 CONTROLLED TEST
MARKS	50
TIME	1.5 hours

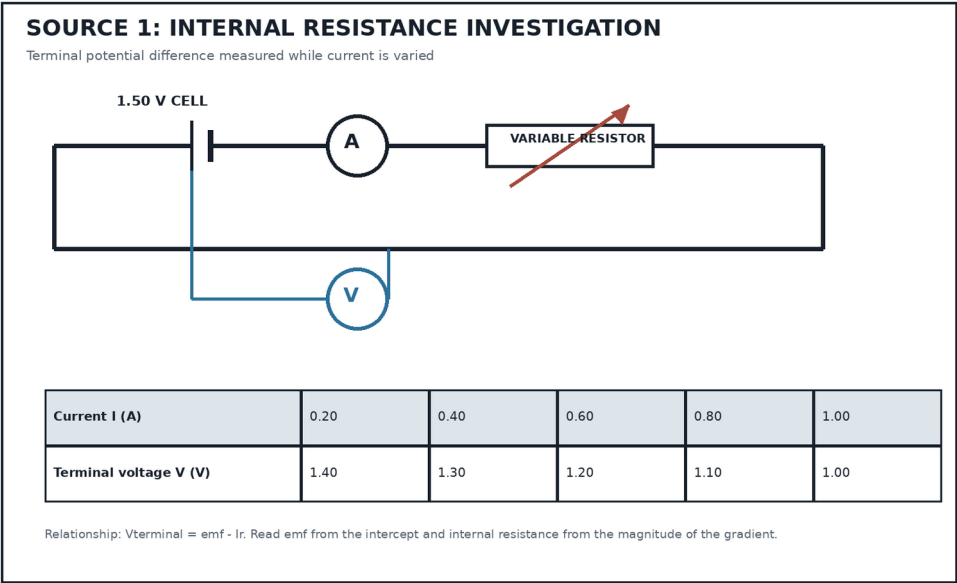
Educator approval required before classroom use.

INSTRUCTIONS AND INFORMATION

- 1. Read all instructions carefully.
- 2. Answer all questions or complete all activities as instructed.
- 3. Show all working where calculations, planning, diagrams or explanations are required.
- 4. Use the space provided by the educator, answer booklet or learner task book.
- 5. Write neatly and clearly.

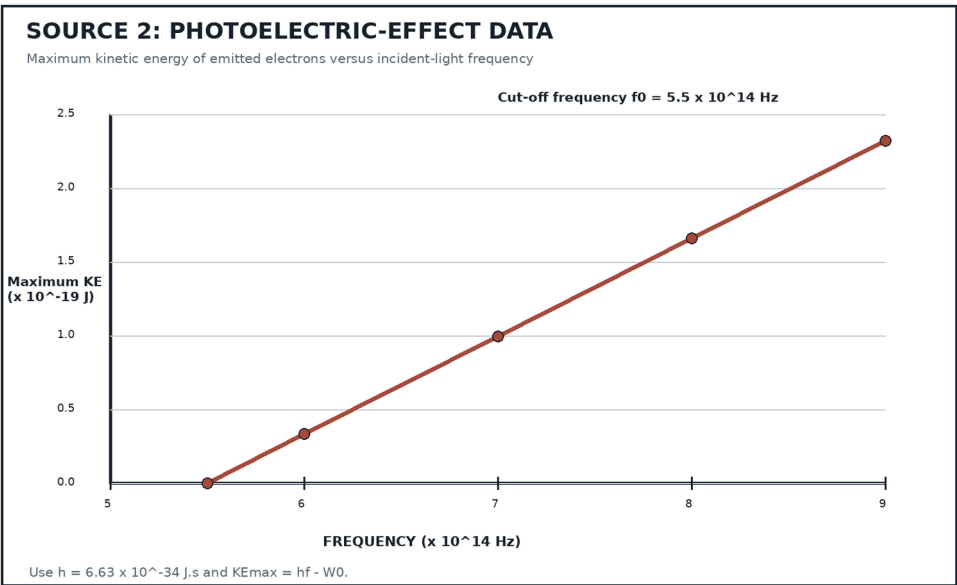
SOURCE MATERIAL

SOURCE 1: Internal resistance investigation



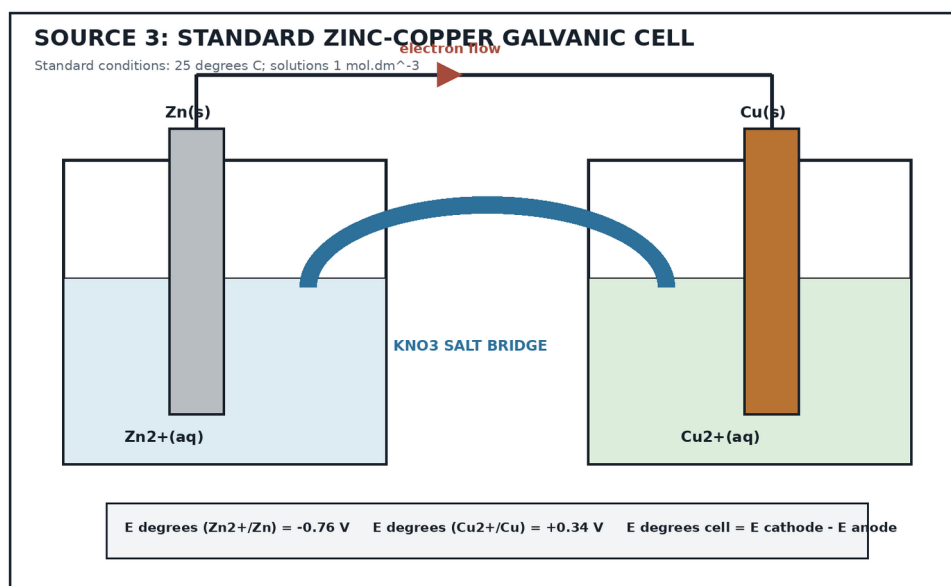
Source Type: Circuit Diagram And Data Table

SOURCE 2: Photoelectric-effect graph



Source Type: Quantitative Graph

SOURCE 3: Zinc-copper galvanic cell



Source Type: Labelled Cell Diagram And Data

QUESTIONS

QUESTION 1: INTERNAL RESISTANCE

Instructions: Use Source 1. Show formulae, substitutions and units.

Stimulus: Refer to SOURCE 1.

1.1. Distinguish between the emf of a cell and its terminal potential difference. (2)

1.2. Identify the independent variable, the dependent variable and ONE controlled variable in this investigation. (3)

1.3. Describe the relationship between terminal voltage and current shown by the data. (2)

1.4. Use the data and $V = \text{emf} - Ir$ to determine the emf and internal resistance of the cell. (5)

1.5. Calculate the terminal voltage and the potential difference across the internal resistance when the current is 0.70 A. (4)

1.6. Explain why the switch should be opened between readings. (2)

QUESTION 2: PHOTOELECTRIC EFFECT

Instructions: Use Source 2 and the supplied constant $h = 6.63 \times 10^{-34}$ J.s.

Stimulus: Refer to SOURCE 2.

2.1. Define the work function of a metal. (2)

2.2. Calculate the work function of the metal. (4)

2.3. Calculate the maximum kinetic energy of emitted electrons when the frequency is 8.0×10^{14} Hz. (4)

2.4. State the effect of increasing light intensity when the frequency is above f_0 , and explain what happens when the frequency is below f_0 . (3)

2.5. State why the photoelectric effect supports the particle model of light. (2)

QUESTION 3: GALVANIC CELLS

Instructions: Use Source 3. Standard conditions apply.

Stimulus: Refer to SOURCE 3.

3.1. Identify the anode and the cathode.

(2)

3.2. State the direction of electron flow in the external circuit.

(2)

3.3. Write the oxidation and reduction half-reactions.

(4)

3.4. Write the overall cell reaction.

(2)

3.5. Calculate the standard emf of the cell and state whether the reaction is spontaneous.

(3)

3.6. Explain the function of the salt bridge.

(2)

3.7. State how the cell potential changes as Zn^{2+} concentration increases and Cu^{2+} concentration decreases while the cell discharges.

(2)
